An Adapted Mathematical Model for Body Composition

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1 Basis of Work

Although the processes of metabolism and biological functions of the human body are quite complex, we are able to mathematically model bodily phenomena due to the fact that the laws of thermodynamics apply to the whole-body system. The first law of thermodynamics states that "the total energy of an isolated system remains constant; it is said to be conserved over time."[1] With regard to the human body, fluctuations in the body's energy content are simply due to imbalances between one's intake and expenditure of energy. Furthermore, because energy is stored as either lean mass or fat mass, we can predict changes in one's body mass and composition through an energy balance model. The one that I found to be much more useful and encompassing is one that was developed by a Kevin D. Hall, who has made major contributions with regard to mathematically modelling human biology, and his model consists of 5 differential equations.

1.1 Glycogen storage

The first of these differential equations pertains to the intake of energy through food. Glucose is the most important energy fuel for all organisms, and circulates within the human body as blood sugar. Excess energy, which is in surplus for those who are obese, that is consumed from food is stored in glycogen reserves. Glycogen is primarily found in one's liver and muscle tissue, and glycogen content is mostly dependent on carbohydrate intake (CI). Below is the differential equation for the change in glycogen over time. [3]

$$\rho_G \frac{dG}{dt} = CI - k_G G^2 \tag{1}$$

When $\frac{dG}{dt} = 0$ we say that G is at an equilibrium, meaning that it will stay constant. At this equilibrium, the glycogen term is a quadratic, meaning that the carbohydrate intake must be increased four-fold to increase glycogen by a factor of 2, and the parameter ρ_G=0.004 kcal/kg represents the energy density

of carbohydrate.[2] Glycogen is important for the change in fat and lean mass, as it can be thought of as storing excess energy consumed through food and thus increasing the overall body fat% of an individual. The more glycogen content a person has directly means they will have more stored energy, and thus a larger amount of fat mass. This is why, as seen in the equation, glycogen content is mainly dependent on carbohydrate intake. When the body consumes carbohydrates, glycogen increases until the quadratic term catches up, causing the body reach a new equilibrium in glycogen. To calculate the unknown constant k_G, I simply set the derivative equal to zero and solved for k_G.

$$\frac{dG}{dt} = 0 \tag{2}$$

$$k_G = \frac{CI(0)}{G(0)^2} \tag{3}$$

1.2 Extracellular Fluid

Extracellular fluid, which describes the bodily fluid outside the cells of the human body. This mainly accounts for the water that is retained in the body, and changes to this are dependent on changes to one's sodium intake. Although this was part of Hall's energy balance model, I will not include this in my adapted mathematical model for numerous reasons. Firstly, extracellular fluid contributes to one's body weight but has no affect on one's body composition, which is what is being explored in this investigation. Secondly, changes in extracellular fluid are heavily dependent on one's CI, which is kept constant in my model. Because of this, I disregarded this aspect of Halls energy balance model.

1.3 Energy Partitioning

Energy within the body is stored as either lean or fat tissue. As stated previously, changes in the proportion of lean mass to body fat depends on one's energy intake and expenditure. This is modeled in the two equations below [3]

$$\rho_F \frac{dF}{dt} = (1-p)(EI - EE - \rho_G \frac{dG}{dt}) \tag{4}$$

$$\rho_L \frac{dL}{dt} = p(EI - EE - \rho_G \frac{dG}{dt}) \tag{5}$$

Where ρ_F is the energy density per unit change of fat and ρ_L is the energy density per unit change of lean mass. p here serves as a dimensionless energy partition function, which in the context of this equation simply expresses the way in which the body partitions excess energy between lean mass and fat mass after the rate of change of glycogen has been accounted for and subtracted out. Dimensionless just means that this value is a scalar, and will result in a real number. The function can be expressed as $p = \frac{C}{C+F}$ with C being a constant. p is extremely important here, as it marks the only difference between these two

equations. p will always be a fraction, and thus determines the proportion of excess energy that will become lean tissue. Energy expenditure (EE) is seen in the equation

$$EE = \frac{K + \gamma_F F + \gamma_L L + \delta BW + TEF + AT + (EI - \rho_G \frac{dG}{dt}[p\frac{\eta_L}{\rho_L} + (1-p)\frac{\eta_F}{\rho_F}]}{1 + p\frac{\eta_L}{\rho_L} + (1-p)\frac{\eta_F}{\rho_F}} \tag{6}$$

The equation for Energy expenditure is quite complicated with a variety of variables, however when evaluated I was able to understand the work at play. K here is a calculated constant, which I was able to find using the same method when I found k_G, and this is expressed below

$$EE(0) = EI(0) \tag{7}$$

$$K = EI(0) - \gamma_F F - \gamma_L L - \delta BW \tag{8}$$

Furthermore, $\gamma_F F$= 3.2 kcal $(kg^-1)(d^-1)$ and $\gamma_L L$= 22 kcal$(kg^{-1})(d^{-1})$ and are regression coefficients that describe the contributions of both fat mass and lean mass to resting metabolic rate, and η_L and η_F represent the energy that is expended to change fat mass and lean mass. [2] As stated earlier, p is a partitioning function that takes into account the rate of change of glycogen. However, when looking at energy expenditure, energy is expended to change fat mass and lean mass, which is why this change is not only multiplied by both p and $1-p$ for lean mass and fat mass respectively, but also by the rate of change of glycogen. It was assumed that the energy expenditure of most physical activities is proportional to body weight and is thus specified by the parameter δ. TEF, which is the thermic effect of feeding, is $= .1\Delta EI$, and will vary based on the amount of food digested, as the energy expended on digestion is dependent on the amount needed to digest, and thus energy intake as seen in the equation above.[2] This algebraic expression was incredibly difficult for me to understand, but I was able to grasp a baseline understanding of the relationship between the key terms and why it is important to understand how EE is calculated when simulating these differential equations.
Energy expenditure through physical activity can be modeled through the equation below [3]

$$\delta = \frac{[(1-\beta_{TEF})PAL - 1]RMR}{BW} \tag{9}$$

where the physical activity level (PAL), body weight (BW), and resting metabolic rate (RMR) are all factors that impact the amount of energy expended due to physical activity.

Resting metabolic rate (RMR) was calculated in this by Hall through Mifflin St. Jeor's equation which is seen below. [3]

$$RMR = 10 * Weight(kg) + 6.25 * Height(cm) - 5 * Age(y) + 5 \tag{10}$$

However, in my mathematical model, I will be using Katch-McArdel's formula, where [4]

$$RMR = 21.6L + 37 \tag{11}$$

as it is directly dependent on the amount of lean mass.

1.4 Adaptive Thermogenesis

Adaptive Thermogenesis (AT) is defined as the decrease in energy expenditure beyond what could be predicted from body weight or its components (fat-free mass and fat mass) in response to a decrease in energy intake. AT is in opposition of weight change and brings EE into equilbrium with EI. An example that helped me visualize this phenomenon was to think about the opposing relationship between friction and the movement of a pendulum, and AT and weight change. AT is expressed in the following differential equation. [3]

$$\tau_{AT}\frac{dAT}{dt} = \beta_{AT}\Delta EI - AT \tag{12}$$

where β_{AT} is simply a parameter for AT and is equal to .14, and τ_{AT} is equal to 14 days, which is the estimated time constant for the beginning of AT. [3] This is particularly relevant when the diet of an individual is changing, but we assume for the purposes of this model that energy intake is kept constant, which makes the effects of AT minimal.

2 Impact of Physical Activity

When simulating and comparing my adapted Hall model with the data web simulator, I wished to examine the effects of resistance training (RT) and endurance training (ET). Resistance training is the incorporation of weights, which can include one's own body weight, when performing anaerobic exercises, with the purpose of producing muscular contractions to increase one's strength. Endurance training describes the the mode of exercises that increases ones capacity for respiratory control through cardiovascular training, with examples being running and cycling.

2.1 Effects of RT and ET on Lean and Fat Mass

I personally have been able to integrate both forms of training in my life, both seen through my high commitment to basketball (ET) and weight lifting (RT). It is quite intriguing to me to mathematically explain the effects these have on body composition, and to see what is the most effective way to improve my proportion of fat versus lean mass. Physical activity through endurance training, which for me is in the form of basketball, causes weight loss due to an increase in energy expenditure. The way in which we describe this effect is through a physical activity level scale (PAL) which was used and seen in both the web based simulator and Hall's energy balance model, where my estimated PAL value of 1.8 was used. Physical activity through RT triggers many responses in the body, ranging from hormonal changes and growth factors that in short lead to an increase in muscle mass. From personal experience, I noticed that

the rate of muscle hypertrophy (growth) was at its highest when I first began weight lifting seriously last year for the first two months of working out, and ever since has followed a much more steady growth path.

2.2 Mathematical Model

I was able to examine the effect of ET on EE and modeled it by simply increasing the PAL value. However, Hall's equation for the change in lean mass over time did not include a response to RT. Because of this, I added a Gaussian esq function to the this differential equation that resulted in the following model for total change in lean mass over time.

$$\frac{dL}{dt} = p(EI - EE - \rho_G \frac{dG}{dt}) + (.25)re^{-(L-60)^2/200} \tag{13}$$

The reason for adding this term in the above expression is to describe the impact of RT on lean muscle growth, which the hall model failed to encapsulate. I was inspired by the Gaussian (also known as normal) distribution as the shape of this curve is very similar to the predicted shape of the effect of resistance training on the creation of lean mass, as the highest rate of muscle gain is seen in the first 6-15 weeks and is greatly dependent on the initial lean mass of an individual. For an example, someone with a very large amount of lean mass will find it much more difficult to increase this in any significant way, whereas someone with relatively less lean mass has more room to grow. The formula for the curve of a normal distribution is given below

$$f(x) = ae^{-(x-b)^2/2c^2} \tag{14}$$

where a, b and c are all real constants. When we draw a graph for this function, we get a bell-shaped curve which is symmetric at the center. The parameter a defines the highest peak of the curve, b denotes the location of center of this peak, and c is the standard deviation, and controls the width of the curve. I chose 60 as my value for b, as this is roughly the average amount of lean mass in the average male. My denominator came out to be 200 as after trial and error, a vale of 10 for c was the best fit for reaching the desired shape for this function. Similarly, .25 was a computational value that controlled the peak of the curve, and the parameter r is a arbitrary value that varies based on higher frequency, volume, or intensity of training.

3 Validation

In order to examine if this model is accurate, I decided to compare this to a 8 month study that examined the bodily impacts of aerobic, anaerobic, and both at the same time on a group of randomly selected individuals. [5] The relatively close agreement between the model predictions and the processed data for each of the studied groups provides some validation for this model. A comparison of the two outcomes following 8 months of strictly resistance training are shown

below. The data used for the experimental group were mean values of the group where n=44. My r value was set to be relatively high at .0195, which although seems quite low, has a large impact on the results of the model. Similarly, I did the same thing for ET, where PAL is simply changed and there is no term for RT. Since the experimental group of n=38 performed an average of 17 minutes per day of cardiovascular activity, it was discovered that this light to moderate activity level would result in PAL level of 1.5526 and would be close to the mean experimental value. Lastly, for the group for both RT and ET where n=37, I used the updated ET PAL value along with my RT term, and with a new r value of .01826 to encapsulate the intensity of the work at hand. The following tables and graphs for all three groups are seen below

Variable	Baseline	Experimental Change (Std. Dev)	Predicted Change
Lean Mass (Kg)	53.3	-.1 (1.22)	-.2368
Fat Mass (Kg)	34.7	-1.66 (2.67)	-.7815

Table 1: ET validation

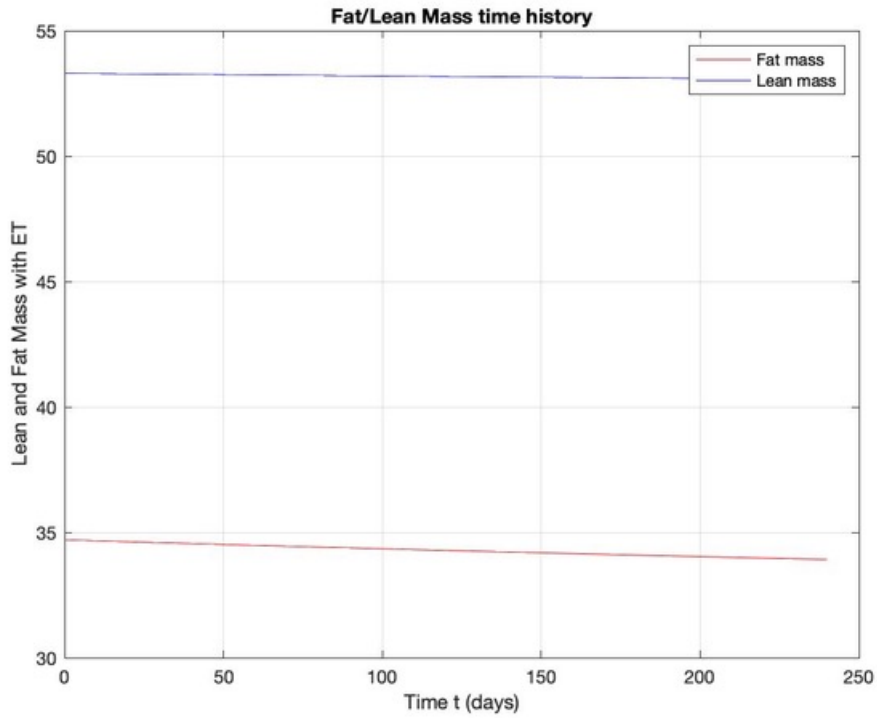
Variable	Baseline	Experimental Change (Std. Dev)	Predicted Change
Lean Mass (Kg)	54.4	1.09 (1.54)	.9549
Fat Mass (Kg)	34.3	-.26 (2.16)	-.2713

Table 2: RT validation

Variable	Baseline	Experimental Change (Std. Dev)	Predicted Change
Lean Mass (Kg)	54	.81(1.38)	.4566
Fat Mass (Kg)	34.9	-2.44(2.97)	-1.5464

Table 3: RT and ET combined Validation

As seen in the table, the predicted change that was found through my adapted mathematical model was relatively accurate to the results that were yielded in an actual experimental setting. All predicted change values fell within a standard deviation of the actual change, and although my values lacked the precision that was initially desired, it was able to correctly predict the trends and general increments in which these changes would occur. Below are visuals that graphically represent these predicted trends.


Fat/Lean Mass time history
Fat mass
Lean mass
Lean and Fat Mass with ET
Time t (days)
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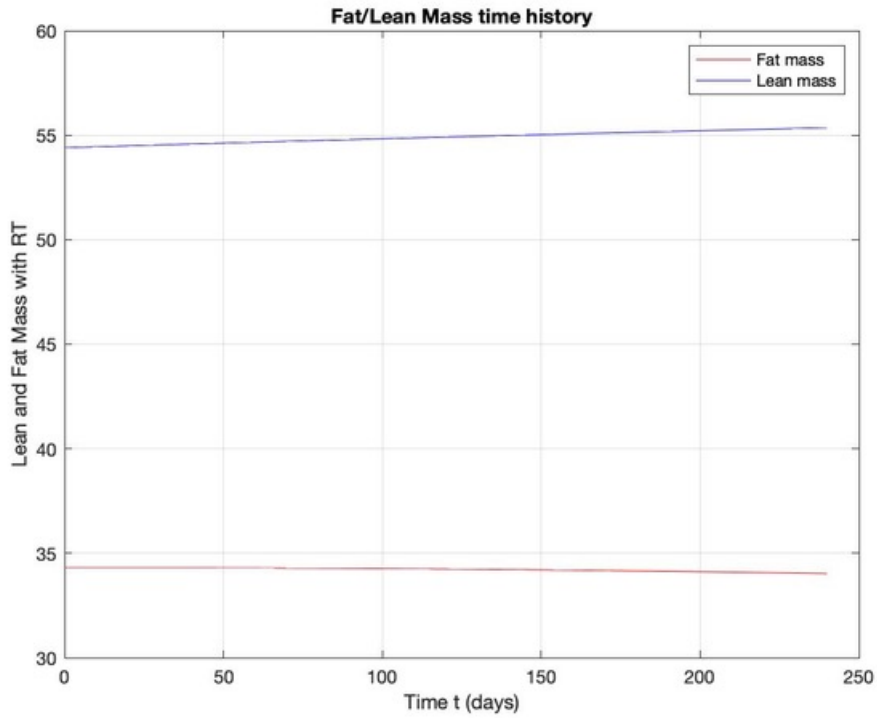
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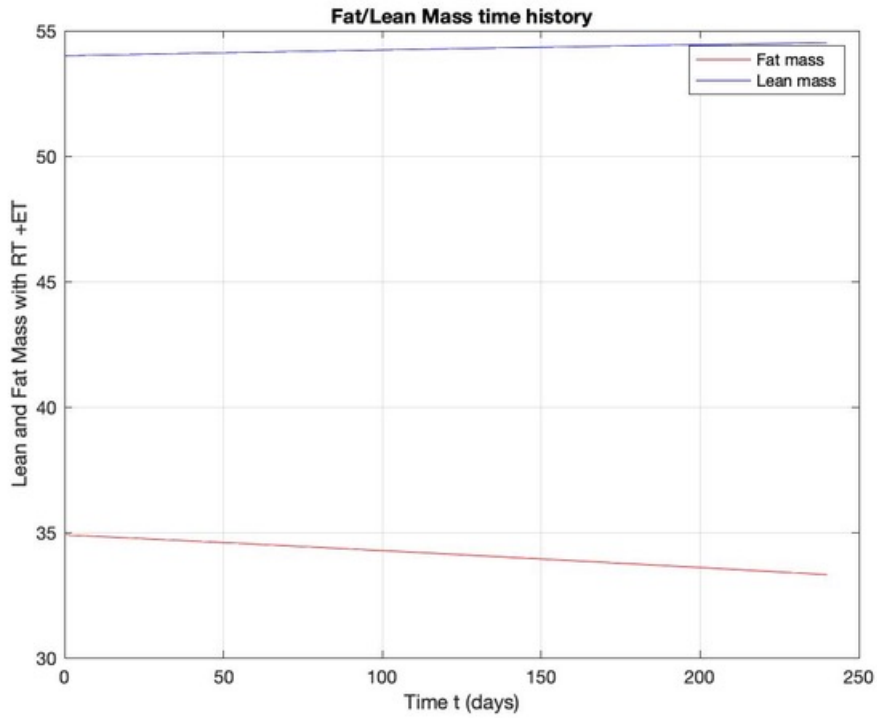

As seen in the graphs above, there is truly only one group that was able to both effectively lose fat mass and gain lean mass simultaneously over the studied 240 day time frame. Firstly, the ET group saw barely any change with regard to lean mass, with the only small change being in fact a negative one. However, there was a larger change in the amount of fat mass, which was lost meaningfully. On the flip side, the RT group saw a different effect, where there was minimal change in the amount of fat mass, with the little change being negative, and saw a large increase in the amount of lean mass generated. In order to reach the best value for body composition, lean mass needs to be increasing and fat mass needs to be decreasing. Although this was seen in the RT group, the graphs and table express that the RT + ET group were able to both gain a large amount of lean mass and lose a large amount of fat mass. As seen above, the graph for the RT + ET group is the only one where both lines have a clear and identifiable slope in the opposing directions, and the data of a .4566 predicted change in lean mass and a -1.5464 change in fat mass express this as well.

References

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[3] Torres, Marcella, Eric T. Trexler, Abbie E. Smith-Ryan, and Angela Reynolds. "A Mathematical Model of the Effects of Resistance Exercise-induced Muscle Hypertrophy on Body Composition." SpringerLink. December 18, 2017. Accessed June 21, 2019. https://link.springer.com/article/10.1007/s00421-017-3787-6.

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